

ChE 205 – Computational Methods in Chemical Engineering – Fall 2019

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem # 1c, 2, 3a, and 4)

Application of computational methods in solving chemical engineering problems.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 1c, 7 pts	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (<3 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (3-5 pts)	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (6-7 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 2, 10 pts	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (<4 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (4-7 pts)	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (8-10 pts)
4. Translates academic theory into engineering applications Problem 4, 5 pts	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (< 2 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (2-3 pts)	Does not appear to grasp the connection between theory and the problem (4-5 pts)
6. The uses of appropriate resources needed to solve problems Problem 3a, 3 pts	Uses no resources to solve problems (< 2 pts)	Uses limited resources to solve problems (2 pts)	Uses appropriate resources to locate information needed to solve problems (3 pts)

Averages:

Performance Indicators: (1) 2.53 (2) 1.36 (4) 2.51 (6) 2.42

Overall Average: 2.21

Standard Deviation: 0.49

Average scores were fairly uniform across Performance Indicators (PI) 1, 2, and 6. A rather low score was observed in PI 2, designated for problem 2. Also, there was wide variation of PI among students. This data suggests that most students were strong in (i) applying mathematical and scientific principles to formulate models, (ii) translate

academic theory into engineering applications, and (iii) uses of appropriate resources needed to solve problems; however they were weak in solving engineering problems using differential equations and linear algebra (31/45 were unsatisfactory to solve problem 2).

Problem 1 (10 pts). Consider the following nonlinear equation:

$$a + \frac{b}{x} = \frac{d}{(x - c)}, \quad (1)$$

where $x \geq 1$, $a = 10$, $b = 4.238$, $c = 0.038$, and $d = 42.917$. Find x for which equation 1 is satisfied, using the following steps:

- Plot the function whose root you are trying to find. (2 pts)
- Based on your results in part a, make a good initial guess, x_0 . (1 pt)
- Write a VBA script to find the root of the function in part a, using Newton's method, and the initial guess, x_0 , you made in part b. The VBA script should take a , b , c , d , x_0 , and the tolerance, tol , as inputs through the spreadsheet. Then it should output
 - the solution, x ,
 - the total number of iterations, and
 - the error at each iteration, as shown below. (7 pts)

	A	B	C	D	E	F
1	10 =a				iteration	error
2	4.238 =b				1	???
3	0.038 =c				2	???
4	42.917 =d				3	???
5	1.00E-09 =tolerance				4	???
6	??? =x0				5	???
7	??? =x				6	???
8	??? =iterations				7	???
9					8	???
10						

Figure 1: Sample input and output of VBA script in Problem 1.

Problem 2 (10 pts). Using the Crank-Nicholson method, write a VBA program to solve the following diffusion equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - B \frac{\partial u}{\partial x}, \quad (2)$$

subject to the following initial condition:

$$u(x, t = 0) = \sin(\pi x), \quad (3)$$

and boundary conditions:

$$u(x, t) = 0 \text{ at } x = 0 \quad (4)$$

$$\frac{\partial u(x, t)}{\partial x} = 1 \text{ at } x = 1, \quad (5)$$

Set $B = 1$, and use $t_{max}=1$, $\Delta x=0.05$, and $\Delta t=0.001$ to plot $u(x, t)$ against $0 \leq x \leq 1$ for four different values of $t=0$ (blue), 0.01 (green), 0.1 (red), and 1 (black).

Problem 3 (10 pts). A model for a batch reactor is given by:

$$\begin{aligned}\frac{dy}{dt} &= b_1 y - \frac{b_1}{b_2} y^2, & y(t=0) &= 0.03 \\ \frac{dz}{dt} &= b_3 y, & z(t=0) &= 0.0\end{aligned}$$

where

y = dimensionless concentration of the first component

z = dimensionless concentration of the second component

t = dimensionless time, $0 \leq t \leq 1$.

Experiments have determined that $b_1 = 13.1$, $b_2 = 0.94$, and $b_3 = 1.71$.

- (a) Solve this ordinary differential equation - initial value problem (ODE-IVP) using forward Euler in spreadsheet (this is not VBA programming) with $\Delta t = 0.005$. Plot y and z in red and blue, respectively, vs $0 \leq t \leq 1$. (3 pts)

Problem 4 (5 pts). A heterogeneous reaction is known to occur at a rate described by the following expression:

$$\log_{10}(r) = \log_{10}(k_1 P_A) - 2 \log_{10}(1 + k_A P_A + k_B P_B), \quad (6)$$

where $\log_{10}$ is the log base 10, and $k_1 = 0.015$. Using an optimization method and the following rate data, estimate the values of k_A and k_B .

P_A	P_B	r
1	0	3.40E-05
0.9	0.1	3.60E-05
0.8	0.2	3.70E-05
0.7	0.3	3.90E-05
0.6	0.4	4.00E-05

Table 1: Rate data for different values of P_A and P_B .

Solutions

Problem 1c:

Sub Newton()

Dim maxiter, iter, row As Integer

Dim tol, err As Double

Dim a, b, c, d, x0, x, f, fprime As Double

'Make the current Worksheet the selected one:

Worksheets(ActiveSheet.Name).Activate

'Clear up the page

Range("A7:B10").Clear

Range("E2:F20").Clear

maxiter = 100 'Maximum number of iterations

a = Cells(1, 1)

b = Cells(2, 1)

c = Cells(3, 1)

d = Cells(4, 1)

tol = Cells(5, 1)

x0 = Cells(6, 1)

Cells(1, 5) = "iteration"

Cells(1, 6) = "error"

err = 1

iter = 0

x = x0

row = 1

While (Abs(err) > tol And iter < maxiter)

 iter = iter + 1

 Call Fcalc(a, b, c, d, x, f, fprime)

 err = f / fprime

 x = x - err

 row = row + 1

 Cells(row, 5) = iter

 Cells(row, 6) = err

Wend

If (iter >= maxiter) Then

 Cells(6, 1) = "Maximum iteration reached;"

 Cells(7, 1) = "Solution might not be valid"

Else

 Cells(7, 2) = "="&x

 Cells(7, 1) = x

 Cells(8, 2) = "="&iterations

 Cells(8, 1) = iter

End If

Worksheets(ActiveSheet.Name).Cells.Font.Size = 16

Worksheets(ActiveSheet.Name).Cells.Font.Name = "Arial"

Range("F2:F20").NumberFormat = "0.00E+00"

```

End Sub
Sub Fcalc(a, b, c, d, x, f, fprime)
    f = (a + b / x) - d / (x - c)
    fprime = -b / x ^ 2 + d / (x - c) ^ 2
End Sub

```

Problem 2:

Option Explicit

Option Base 1

```

Sub CrankNicholsonB()
Dim Bcoeff, Deltax, Deltat, Ratio As Double
Dim Numxsteps As Long
Dim Maxtime As Double
Dim Time As Double
Dim V() As Double
Dim U() As Double

```

```

Dim C() As Double
Dim D() As Double
Dim E() As Double
Dim B() As Double
Dim X() As Double
Dim i, j As Long

```

```

'clear the active sheet
Dim Lastrow As Long
Lastrow = Range("A" & Rows.Count).End(xlUp).Row
Range("A7:V" & Lastrow).Clear

```

```

Bcoeff = ActiveSheet.Cells(1, 3).Value 'get B coefficient
Deltat = ActiveSheet.Cells(2, 3).Value 'get time step
Deltax = ActiveSheet.Cells(3, 3).Value 'get distance step
Ratio = Deltax ^ 2 / Deltat

```

```

Maxtime = ActiveSheet.Cells(2, 5).Value 'get maximum time
Numxsteps = Round(1 / Deltax) - 1
Time = 0 'initialize time

```

```

ReDim V(1 To Numxsteps + 2) As Double
ReDim U(1 To Numxsteps + 2) As Double

```

```

ReDim C(1 To Numxsteps) As Double 'lower diagonal coefficients
ReDim D(1 To Numxsteps) As Double 'diagonal coefficients
ReDim E(1 To Numxsteps) As Double 'upper diagonal coefficients
ReDim B(1 To Numxsteps) As Double 'right hand side vector
ReDim X(1 To Numxsteps) As Double 'unknown temperatures at new time step

```

```

'Get the initial temperatures from the spreadsheet

```

```

i = 6

```

```

For j = 1 To Numxsteps + 2
    V(j) = ActiveSheet.Cells(i, j + 1)
Next j

```

```

Time = 0
While Time <= Maxtime

'Set up the tridiagonal system coefficients

For j = 2 To Numxsteps + 1
    C(j - 1) = -1 - Bcoeff * Deltax / 2
    D(j - 1) = 2 * (1 + Ratio)
    E(j - 1) = -1 + Bcoeff * Deltax / 2
    B(j - 1) = (1 + Bcoeff * Deltax / 2) * V(j - 1) - 2 * (1 - Ratio) * V(j) + (1 - Bcoeff * Deltax / 2) * V(j + 1)
Next j

'right boundary correction
D(Numxsteps) = D(Numxsteps) + 4 / 3 * E(Numxsteps)
C(Numxsteps) = C(Numxsteps) - 1 / 3 * E(Numxsteps)
B(Numxsteps) = B(Numxsteps) - 2 * Deltax / 3 * E(Numxsteps)

Call Thomas(Numxsteps, C, D, E, B, X)

U(1) = 0
For j = 1 To Numxsteps
    U(j + 1) = X(j)
Next j
U(Numxsteps + 2) = (4 / 3) * U(Numxsteps + 1) - (1 / 3) * U(Numxsteps) + 2 * Deltax / 3

Time = Time + Deltat 'this is the "new" time

'display the temperatures at the new time
i = i + 1 'move down one row
ActiveSheet.Cells(i, 1) = Time
For j = 1 To Numxsteps + 2
    ActiveSheet.Cells(i, j + 1) = U(j)
Next j

For j = 1 To Numxsteps + 2
    V(j) = U(j)
Next j
Wend

'change worksheet font size and name
With ActiveSheet
    .Cells.Font.Size = "12"
    .Cells.Font.Name = "Arial"
End With

'set the number of decimal digits
Columns(1).NumberFormat = "0.000"
Range("B5:V" & Lastrow).NumberFormat = "0.00"

End Sub

Sub Thomas(N, C, D, E, B, X)
Dim Beta() As Double
Dim Gamma() As Double
ReDim Beta(1 To N) As Double

```

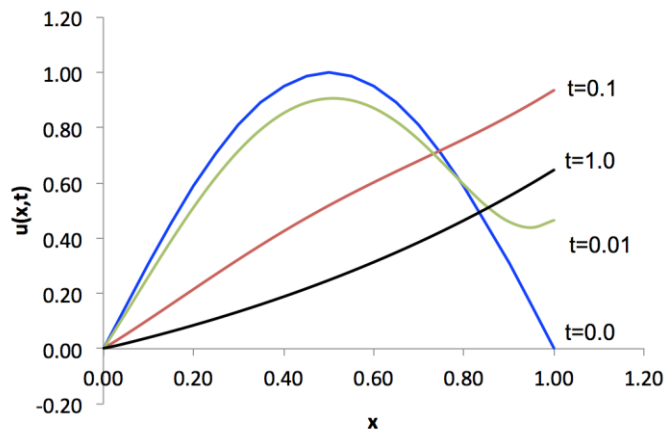
```

ReDim Gamma(1 To N) As Double
Dim i As Long
Dim j As Long

Beta(1) = D(1)
Gamma(1) = B(1) / Beta(1)
For i = 2 To N
    Beta(i) = D(i) - C(i) * E(i - 1) / Beta(i - 1)
    Gamma(i) = (B(i) - C(i) * Gamma(i - 1)) / Beta(i)
Next i
X(N) = Gamma(N)
For i = 2 To N
    j = N - i + 1
    X(j) = Gamma(j) - E(j) * X(j + 1) / Beta(j)
Next i

End Sub

```



Problem 3a

Option Base 1

Option Explicit

Sub rk2and4()

'Main routine for solving a set of ODE's using the RK2 Method.

```

Dim y() As Double      'size N
Dim f() As Double      'size N
Dim h As Double        'fixed time step
Dim b1, b2, b3 As Double 'intermediate time
Dim TimeMax As Double  'maximum integration time
Dim Time As Double     'time counter
Dim ActRow As Integer  'row counter
Dim i As Integer       'a counter
Dim N As Integer       'N = number of state variables and equations

```

```

Dim w() As Double      'size N
Dim k1(), k2(), k3(), k4() As Double 'size N

```

Worksheets(ActiveSheet.Name).Activate

'clear the worksheet

Dim Lastrow As Long


```

Lastrow = Range("A" & Rows.Count).End(xlUp).Row
If Lastrow > 9 Then
    Range("E9:G" & Lastrow).Clear
End If

N = 2          'The number of ODEs
ReDim y(N) As Double          'size N
ReDim f(N) As Double          'size N
ReDim w(N) As Double          'size N
ReDim k1(N), k2(N), k3(N), k4(N) As Double 'size N

TimeMax = ActiveSheet.Cells(1, 2).Value 'get TimMax from spreadsheet
h = ActiveSheet.Cells(2, 2).Value       'get DeltaT from spreadsheet
b1 = ActiveSheet.Cells(3, 2).Value      'get coefficient b1
b2 = ActiveSheet.Cells(4, 2).Value      'get coefficient b2
b3 = ActiveSheet.Cells(5, 2).Value      'get coefficient b3

For i = 1 To N
    y(i) = ActiveSheet.Cells(8, i + 5).Value 'get initial y values
Next i

ActRow = 9          'calculated output starts here
Time = 0
Call FCalc(b1, b2, b3, y(), f())
While (Time <= TimeMax)
    #####
    ' RK2
    For i = 1 To N
        k1(i) = h * f(i)
        w(i) = y(i) + k1(i)
    Next i
    Call FCalc(b1, b2, b3, w(), f())
    For i = 1 To N
        k2(i) = h * f(i)
        y(i) = y(i) + 1 / 2 * (k1(i) + k2(i))
    Next i
    #####
    ' RK4
    For i = 1 To N
        ' k1(i) = h * f(i)
        ' w(i) = y(i) + k1(i) / 2
    Next i
    'Call FCalc(b1, b2, b3, w(), f())
    For i = 1 To N
        ' k2(i) = h * f(i)
        ' w(i) = y(i) + k2(i) / 2
    Next i
    'Call FCalc(b1, b2, b3, w(), f())
    For i = 1 To N
        ' k3(i) = h * f(i)
        ' w(i) = y(i) + k3(i)
    Next i
    'Call FCalc(b1, b2, b3, w(), f())
    For i = 1 To N
        ' k4(i) = h * f(i)
        ' y(i) = y(i) + (k1(i) + 2 * k2(i) + 2 * k3(i) + k4(i)) / 6
    Next i

```

```

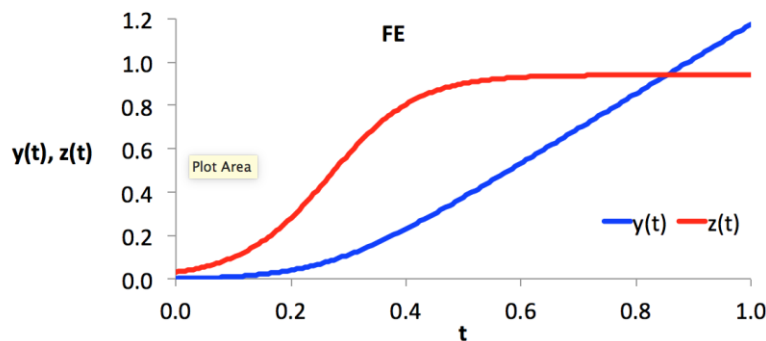
'Next i
'#####

Time = Time + h      'Display the results on the spreadsheet
ActiveSheet.Cells(ActRow, 5) = Time      'put time on spreadsheet
For i = 1 To N
    ActiveSheet.Cells(ActRow, i + 5).Value = y(i) 'output
Next i
ActRow = ActRow + 1
Wend                'repeat for all times
'change the fontsize
Lastrow = Range("E" & Rows.Count).End(xlUp).Row
Range("E6:G" & Lastrow).Font.Size = 14
Range("E6:G" & Lastrow).Font.Name = "Arial"
Range("E8:E" & Lastrow).NumberFormat = "0.000"
Range("F8:G" & Lastrow).NumberFormat = "0.00"
End Sub
'#####
Sub FCalc(b1, b2, b3, w, f)

    f(1) = b1 * w(1) - b1 / b2 * w(1) * w(1)
    f(2) = b3 * w(1)

End Sub
'#####

```



Problem 4

	A	B	C	D	E	F
1	kA=	19.94				
2	kB=	5.01				
3	k1=	0.015				
4						
5	P _A	P _B	r	rcalc	residual	residual^2
6	1	0	3.40E-05	3.42E-05	2.57E-03	0.000
7	0.9	0.1	3.60E-05	3.57E-05	-3.77E-03	0.000
8	0.8	0.2	3.70E-05	3.72E-05	2.55E-03	0.000
9	0.7	0.3	3.90E-05	3.87E-05	-2.90E-03	0.000
10	0.6	0.4	4.00E-05	4.02E-05	1.73E-03	0.000
11						
12	sum					0.000
13						

